Deep Learning on CPU

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Deep Learning on CPUs

CPU for DL in Wearable Systems and Data Centers

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Contents

- CPU core design for wearable systems (e.g., smartwatch).
- Training and inference of ML models on CPUs in data centers.

ML—machine learning

Designing a core for wearable workloads

We will review this paper: S. Mehta and J. Torrellas, "WearCore: A core for wearable workloads," in *Proceedings of the International Conference on Parallel Architectures and Compilation*, 2016, pp, 153–164.

Image source: https://www.wareable.com/wearable-tech/decade-in-wearable-tech-7833

Motivation

- Typical battery capacities: smartphone: 2550 mAh; smartwatch: 300 mAh.
- Making right choice of core type, core count, and their capabilities are essential.
- They study characteristics of wearables to find the optimal core design for them.
- They identify 10 frequently used and computation-intensive applications running on wearables, which have strict QoS requirements.
- These applications, termed WearBench, come from domains such as image/video/audio/speech processing.

QoS—quality of service, mAh—milliampere hour

Apps in WearBench

1. Speech-to-text
2. Part-of-speech (POS) tagging
3. Word chunking
4. Name entity recognition
5. JPEG compression
6. Image denoising
7. Face recognition
8. Image classification
9. MP3 player
10. H264 decoder for video rendering

Characteristics of WearBench Applications

- Most of these applications are parallel and can benefit from using multiple cores.
- They are data/computation-intensive, but not control-intensive.
- They have high L1 data cache miss rate.
- In total, 25 percent of their operations are vector operations, and hence, cache misses have a crucial impact on performance.
- Based on these considerations, they design the cores for wearables.

DSPs: Pros and Cons

- DSPs have VLIW cores for audio/video processing.
- Recent DSPs, for example, Qualcomm's Hexagon DSPs, use three threads, and thus can tolerate memory latency and handle conditional code.
- DSPs do not support multiprocessing in hardware or software, that is, a DSP uses only one core.
  - There is not much inherent parallelism in traditional DSP workloads.
- DSPs also do not support fine-grained parallelism.
- Parallelism is important for minimizing latency of apps in WearBench. Hence, a DSP is not preferred.

DSP—digital signal processor, VLIW—very long instruction word

CPU Core Design for WearBench

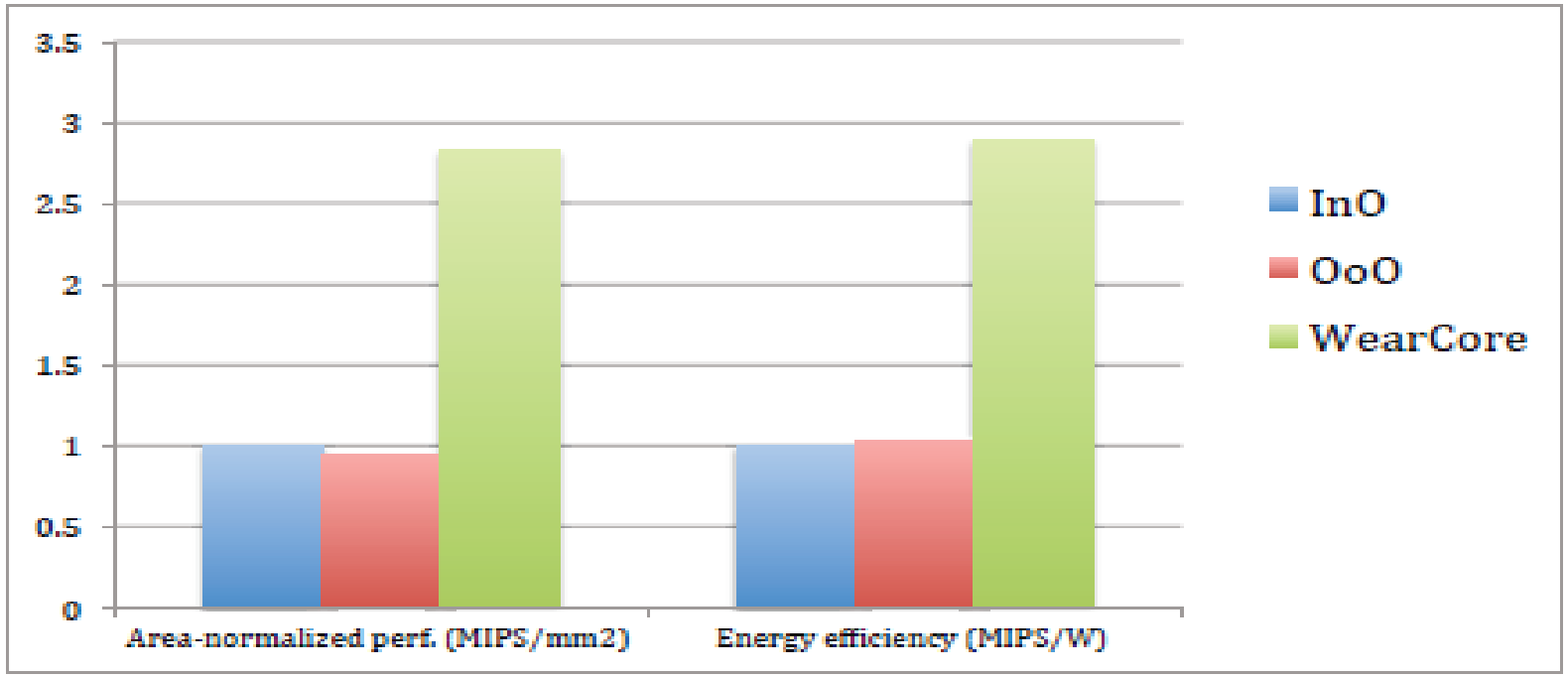
- A smartwatch needs to use multiple (e.g., four) simple cores to exploit parallelism.
- An in-order core is stalled on each miss and WearBench apps have high data cache misses.
- Hence, they use a partial out-of-order core, which
  - avoids read-after-write hazards by using a scoreboard and
  - can have at most two outstanding cache misses.
- It is less complicated than a full out-of-order core
  - due to not using speculation or renaming and
  - having a smaller L2 cache.

Improving Performance

- Scoreboarding is not enough for reaching the performance of an out-of-order core.
- Hence, they use a “software-assisted hardware prefetcher” for prefetching the data to the L1 cache.
  - It reduces the number of L1 cache misses.
- An app developer inserts a prefetching instruction in the program. It specifies which data to prefetch.
- Hardware finds the correct prefetch distance.
- Results: this prefetching technique is crucial for performance and is better than hardware prefetching.

Results

- On the metric of performance/area and performance/power, their core design is better than both in-order and out-of-order cores. Here InO is Cortex-A7 and OoO is Cortex-A15.



S. Mehta et al., "WearCore: A core for wearable workloads," PACT, 2016

Hardware Used by Facebook Data Centers for Supporting Many Machine Learning-Based Services/Products

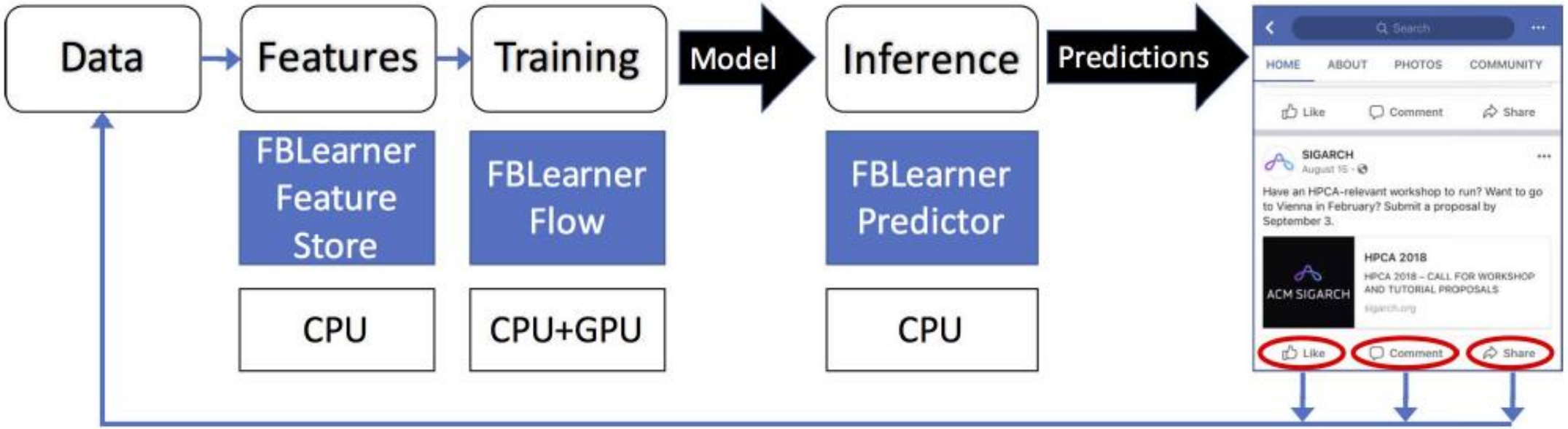
This discussion is based on the following paper: Hazelwood, et al. "Applied machine learning at facebook: A datacenter infrastructure perspective." HPCA, 2018. The images and tables are courtesy of this paper.

ML Models Power Many Facebook Services

Machine learning algorithms leveraged by product/service

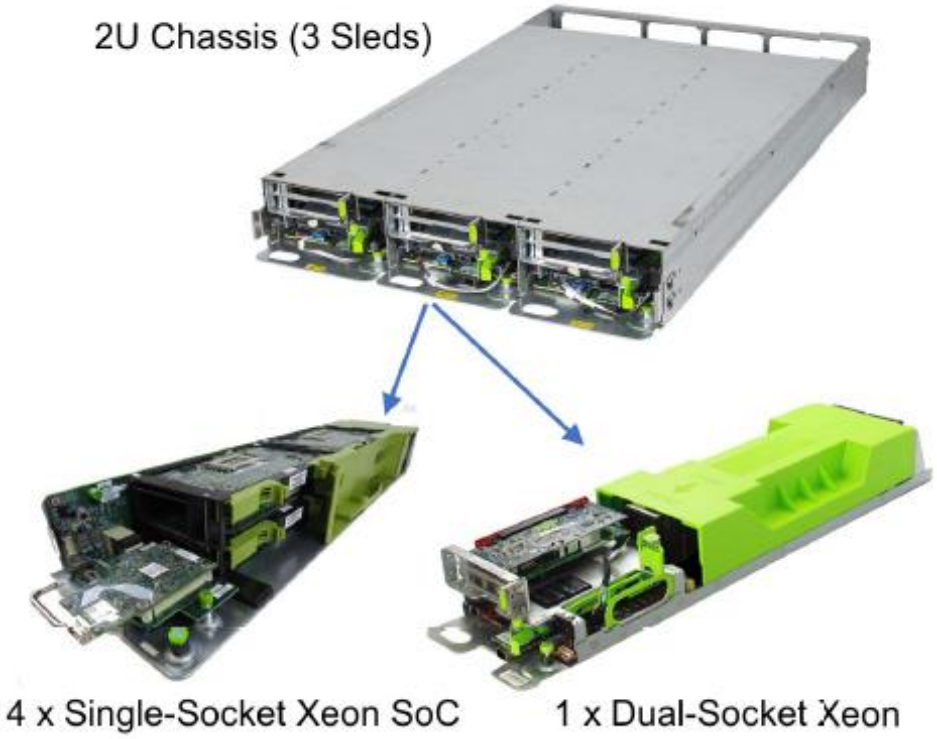
Models	Services
Support Vector Machines (SVM)	Facer (User Matching)
Gradient Boosted Decision Trees (GBDT)	Sigma
Multi-Layer Perceptron (MLP)	Ads, News Feed, Search, Sigma
Convolutional Neural Networks (CNN)	Lumos, Facer (Feature Extraction)
Recurrent Neural Networks (RNN)	Text Understanding, Translation, Speech Recognition

Facebook's machine learning flow and infrastructure



Their Data Centers Use Two Types of Servers

- Single-socket CPU (1×CPU): it supports web tier. It uses a power-efficient CPU (e.g., Broadwell-D) with relatively small memory and storage capacity.
- Dual-socket CPU (2× Skylake SP or Broadwell-EP). It has high memory capacity for supporting memory and computation-intensive applications.



Offline Training (1/3)

- Sigma (used for classification and anomaly detection) and "News Feed" are trained on CPUs.
- Language translation, speech recognition, and Lumos (used for extracting high-level features from images) are trained on GPUs.
- For search, both CPU and GPU are used for training.

Frequency, duration, and resources used by offline training for various workloads:

Service	Resource	Training Frequency	Training Duration
News Feed	Dual-Socket CPUs	Daily	Many Hours
Facer	GPUs + Single-Socket CPUs	Every N Photos	Few Seconds
Lumos	GPUs	Multi-Monthly	Many Hours
Search	Vertical Dependent	Hourly	Few Hours
Language Translation	GPUs	Weekly	Days
Sigma	Dual-Socket CPUs	Sub-Daily	Few Hours
Speech Recognition	GPUs	Weekly	Many Hours

Offline Training (2/3)

For Facer (face detection and recognition)

- A generic model is trained on GPUs after many months.
- A user-specific model is trained on 1×CPUs when the sufficient number of new images have been generated.

Offline Training (3/3)

- GPU is primarily used for offline training.
- Availability of a large number of CPUs makes them an attractive target for DL, especially during off-peak hours.
- The inability to rapidly and continuously train DL models can have severe consequences
  - such as presenting irrelevant/stale news and ads and
  - not blocking spam and offensive contents.
- They also leverage distributed training techniques for performing CPU–GPU heterogeneous computing.

DL—deep learning

Online Inference (1/2)

- Different services have different memory and computation requirements.
- For example, ads ranking performs multiple rounds of screening using an MLP-like model.
- This model has a sparse embedding layer and, hence, has a high memory footprint.
- Therefore, the later rounds, where the memory footprint becomes even higher, are executed on a different server.

MLP = Multi-Layer Perceptron

Online Inference (2/2)

- Online inference is mostly done on 1×CPUs or 2×CPUs.
- Since 1×CPUs have higher energy and cost efficiency, they are preferred over 2×CPUs.
- Some services can be run on powerful mobile devices of end users, which reduces communication overheads.
- 2×CPUs are required for running some bulky services.

Resource requirements of online inference workloads

Services	Relative Capacity	Compute	Memory
News Feed	100X	Dual-Socket CPU	High
Facer	10X	Single-Socket CPU	Low
Lumos	10X	Single-Socket CPU	Low
Search	10X	Dual-Socket CPU	High
Language Translation	1X	Dual-Socket CPU	High
Sigma	1X	Dual-Socket CPU	High
Speech Recognition	1X	Dual-Socket CPU	High